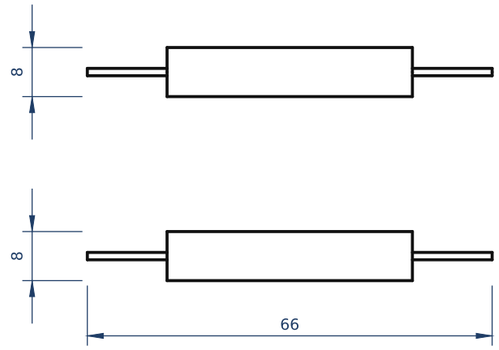
ENVELOPE (as the model draws it) 1:1



specification	value
rating	VRRM 20 kV; IF(AV) 5 mA; IFSM 0.5 A (half-cycle, 50 Hz)
characteristics	VF ≤ 62.5 V at 10 mA; IR ≤ 2 µA at 25 °C, ≤ 5 µA at 100 °C; Cj ≤ 1 pF; trr ≤ 0.1 µs
candidate	XCH 2CL77, LCSC part C2982866 (datasheet: wmsc.lcsc.com …/2204201445_XCH-2CL77_C2982866.pdf)
alternates	Anshan Leadsun 2CL77 (Cj 2 pF, Tj 150 °C); HVGT 2CL77 (100 ns)
envelope	the layout's: Ø8 × 40 body [RH]; confirm on the sheet received
where	D1-D12, one stick each (sim/multiplier-layout-findings.md §2)

NOTES

1 The record's recommended stick (sim/diode-stack-findings.md: 'stock at LCSC; the recommended stick').
2 Datasheets disagree on VF max (62.5 / 75 V) and Tj (130 / 150 °C): design to the sheet of the lot bought.
3 Lot-match the twelve on IR at 15 kV, 25 °C (the clamps and the rings' leakage).

REVISIONS

Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

status	BOUGHT
qty per machine	12
used in	DCK-504
source	sim/diode-stack-findings.md; LCSC C2982866



3-D view (axonometric, not to scale)

DCCREG turbine · bought parts

HV rectifier stick, 20 kV, 5 mA (XCH 2CL77)

Drawing no. **DCCREG-P-002**

Material: avalanche HV silicon rectifier stick, epoxy body

Mass: —	Qty per machine: 12	Rev A Sheet 86 of 103 A3
Scale: 1:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**BOUGHT · SOURCE CONTROL**